Installation Manual

Vehicle Washing Management System (Vehicle Washing)

Requirements, Configuration, Deployment With Docker, SQL Server Database, and Service Verification

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Version Control

Table 1 records the change history of this document. Each time the installation procedure changes (for example, because of a new version of a service or of a dependency), a row must be added with the version, the date, the description, and the person responsible.

Table 1

Version History of the Installation Manual

Version	Date	Description of the change	Responsible
2.0	2026-10-05	Creation of the installation manual (in Spanish) from the project documentation, within version 2.0 of the manual set.	Development team
3.0	2026-10-07	Translation into English, APA Style (7th ed.) formatting, and verification against the infrastructure repository and the code of the services. The placeholders were replaced with the real values: repositories, Compose services, ports, database name, environment variables, migrations per service, payment-service in .NET 8, health endpoints, and demonstration accounts. Corrected: migration strategy (one changelog per service applied at startup, not a single master changelog), absence of account lockout, and gateway behavior (it does not validate tokens).	Development team
3.1	2026-10-09	Update against the code of October 9, 2026, and creation of the Spanish version (Manual_instalacion_LavadoVehicular_3145556_ES.md). Corrected: the operations repository is called *lavarapido-operation-service* and must be cloned with that name; real addresses of the four repositories that were pending; payment-service now runs in Docker with the *app* profile and the gateway container receives ROUTE_PAYMENT_URL; CORS origins include Expo web (8081); new changelogs (security 023, 026, and 030; booking 021, 022, and 024; notification 025, 027, 028, and 029; payment 017 to 020); 49 tables; 22 notification types; payment-service publishes events with MESSAGING_ENABLED; SPRING_PROFILES_ACTIVE added.	Development team
4.0	2026-10-16	Review against the code of October 16, 2026. The figures changed from placeholders to real screenshots of the installation. Updated: 57 tables in the nine schemas (checked with the query in section 11.5), account lockout after failed attempts, the payment-service health endpoint that now checks the database, a reference to the Deployment Document for the Azure server and the APK, and Annex B items that were resolved.	Development team
4.1	2026-10-17	Review against the code of October 17, 2026. Added the optional *EXPO_PUBLIC_WEB_URL* variable of the mobile app (section 10.5) and the native dependencies *expo-image-picker* and *expo-image-manipulator*, which require *npm install* and a new APK (section 13.2).	Development team
4.2	2026-10-19	Continuous integration with GitHub Actions: the scope was updated and it was removed from Annex B.	Development team
4.3	2026-10-20	Review against the code of October 20, 2026: the *notification.notification_type* catalog has 24 rows (migrations 031 and 038 added the alert for a booking without an operator and the new app version notice).	Development team
4.4	2026-10-21	Review against the code of October 21, 2026: optional variables of the gateway session check and of the close of expired bookings (section 10.2), booking-service seeds 110 to 112 (section 11.2), cancellation reasons (Table 9), automatic backups of the public server (section 11.6), check of the token of a closed session (sections 16.3 and 16.4), and new errors (Table 14).	Development team
4.5	2026-10-21	Change of the rule for bookings that end without being closed, reviewed by the team on the night of October 21: booking-service only cancels expired scheduled bookings and operations-service reviews the confirmed and started ones. Added the *FOLLOW_UP_ENABLED* and *FOLLOW_UP_INTERVAL* variables, *BOOKING_EXPIRY_GRACE_MINUTES* shared by booking-service and operations-service, and the services that use *INTERNAL_API_KEY* (section 10.2); changelogs 040 of operations-service and 041 of notification-service (section 11.2); 26 notification types (Table 9); check of the "To close" card (section 16.4); a new error (Table 14) and Annex B.	Development team
4.6	2026-10-23	Review against *develop* of October 23, 2026 and version *v1.0.0*: branches and tags of each repository (8.1); variables the Compose file already passes, including *FOLLOW_UP_LOOK_BACK_DAYS*, *ALLOWED_EMAIL_DOMAINS*, *DEMO_*_PASSWORD*, *ADMIN_DOCUMENT*, and *ADMIN_PHONE*, and *INTERNAL_API_KEY* required for the operator (10.2); the global script *lavarapido-6-services-sqlserver.sql* is not versioned (11.3); the APK used to test push notifications is built with the EAS *preview* profile, because *development* does not exist (13.2); demonstration accounts through variables (15.1). Includes the content of Merari Urbano's *fix/A-05* and *fix/B-02* branches.	Development team
4.7	2026-10-24	Review against *develop* of October 24, 2026, which runs on the public server since that day (8.1); new variables of the gateway (*AUTH_RATE_LIMIT_**, *TRUSTED_PROXIES*, *ROUTE_CONNECT_TIMEOUT*, *ROUTE_READ_TIMEOUT*) and of operations-service (*START_LATE_LIMIT*), *INTERNAL_API_KEY* and *BOOKING_SERVICE_URL* in customer-service, and *BOOKING_EXPIRY_CHECK_** with no effect (10.2); the published web listens on port 8080 (Annex B).	Development team
4.8	2026-10-26	Payment at booking, reviewed against *develop* of October 26, 2026 and deployed to the public server that day: new booking-service variables (*BOOKING_CASH_ACTIVE_LIMIT*, *BOOKING_PAYMENT_**, *PAYMENT_SERVICE_URL*) and *INTERNAL_API_KEY* also in payment-service (Table 7); booking-service migration 049 and seed 117 (Table 8); *booking.payment-events* queue and *RABBITMQ_HOST=127.0.0.1* for payment-service outside Docker on Windows (12.5).	Development team
4.9	2026-10-26	Versions *v1.1.0* (10 repositories, including the first one of the mobile app) and *v1.1.1* (web, mobile app, payment-service, and infra) of October 26, 2026, and recommended tag for an evaluation installation (8.1).	Development team

Version	Date	Description of the change	Responsible
4.10	2026-10-27	Versions *v1.1.1* (customer and booking) and *v1.1.2* (web, mobile app, payment-service, and infra) of October 27, 2026 and tags to use (8.1); 58 tables (11.1, 11.5, and checklist); *operations-service.booking-confirmed* queue in the RabbitMQ topology (12.5).	Development team
4.11	2026-10-27	payment-service *v1.1.3* (approved, pending publication): *DB_PASSWORD* required, RabbitMQ credentials required against a remote RabbitMQ, and local test database *LavaRapidoPaymentIT* with SQL Server 2022 (variables and payment section).	Development team
4.12	2026-11-04	payment-service *v1.1.3* merged into *main* on October 27 and tagged on November 4, 2026: tag to use when cloning (8.1) and payment's test projects (9.3).	Development team

Note. Dates are written in the ISO 8601 format (year-month-day). Prepared by the development team.

Table of Contents

Version Control 2
4. Introduction 6
4.1 Sources of This Document 6
4.2 Scope 6
5. Objective 6
5.1 General Objective 6
5.2 Specific Objectives 6
6. Minimum Hardware Requirements 6
6.1 Supported Operating Systems 7
6.2 Network Ports 7
7. Required Software 7
7.1 Checking the Installed Tools 8
8. Obtaining the Project 9
8.1 Cloning the Repositories 9
8.2 Expected Structure 10
9. Installing Dependencies 10
9.1 Container Images 10
9.2 Java Services 11
9.3 payment-service (.NET 8) 11
9.4 Web Application 11
9.5 Mobile Application 11
10. Configuring Environment Variables 11
10.1 Creating the .env File 12
10.2 Variables of the .env File 12
10.3 Generating Secrets 14
10.4 Example of the .env File 14
10.5 Configuring the Client Applications 14
11. Database Creation and Import 15
11.1 Starting SQL Server and Creating the Database 15
11.2 Migrations of Each Service 16
11.3 Alternative Method: Global Script 17
11.4 Database Accounts 17
11.5 Verifying the Structure and the Initial Data 17
11.6 Backup and Reset of the Local Database 18
12. Running the Backend 18
12.1 Startup Order 18
12.2 Option A: Services in Docker (Recommended) 18
12.3 Option B: Java Services From a Terminal or the IDE 19
12.4 payment-service (.NET 8) Outside Docker 19
12.5 RabbitMQ and Mail 19
12.6 Logs, Stopping, and Updating 20
13. Running the Frontend 20
13.1 Web Application (Angular) 20
13.2 Mobile Application (React Native With Expo) 20
13.3 Static Mockups (Optional) 21
14. Accessing the System 21
14.1 First Access to the Web Application 22
15. Test Users 22
15.1 Demonstration Accounts 22

15.2 Rules for Accounts 22
15.3 Creating a Test Customer 22
15.4 Operator and Administrator Accounts 23
16. Installation Verification 23
16.1 Infrastructure and Database 23
16.2 Messaging and Mail 23
16.3 Services and Gateway 23
16.4 Minimum Functional Flow 24
16.5 Traces Verification 25
17. Common Errors and Solutions 25
18. Glossary 26
References 28
19. Annexes 28
Annex A. Source of the Screenshots 29
Annex B. Information to Confirm Before Delivery 30
Annex C. Summary of Installation Commands 31
Annex D. Security Checklist Before Sharing the Environment 32
Annex E. Delivery Checklist 33

Note. Numbering starts at 4 because sections 1 to 3 are the cover, the version control, and the table of contents, which have no number in their title. It is kept this way so that the cross-references between documents by section number (for example, "Installation Manual, section 10") remain valid.

4. Introduction

The Vehicle Washing Management System (Vehicle Washing), internally called *LavaRapido*, is made up of a gateway (API Gateway), six backend microservices (five in Java 21 with Spring Boot 4 and one in C# with .NET 8), a Microsoft SQL Server 2022 database, a RabbitMQ message broker, a development mail server (Mailpit), and two clients: an Angular 21 web application and an Android application in React Native with Expo. This manual covers the local installation with Docker, which is the one the team uses to develop and test. The public server on Azure, which is turned on when it is needed to present the system, is set up with the scripts in the *deploy* folder of *lavarapido-infra* and is explained in the Deployment Document.

This manual explains how to prepare a machine, obtain the code, configure the environment variables, create the database, start each component in the correct order, and check that the system works. It is addressed to infrastructure and DevOps staff, or to anyone who must start the system on a local machine or a development server. It starts from the premise that an installation must be repeatable, automatable, and verifiable (Humble & Farley, 2010), and that the configuration that changes between environments must reside in the environment and not in the code (Wiggins, 2017).

4.1 Sources of This Document

The procedures were verified against the infrastructure repository (*lavarapido-infra*: *docker-compose.yml*, *.env.example*, and the team guide), the source code of the eleven repositories (*develop* branch as of October 21, 2026), and the project documentation updated on that same date (Vehicle Washing Development Team, 2026a, 2026b). The values that depend on each installation (passwords, keys, and addresses) are written between angle brackets, such as `<SA_PASSWORD>`; no placeholder must be copied literally into a terminal. The implementation of a release in its target environment (entry criteria, acceptance, training, and rollback) is described in the Implementation Procedure.

4.2 Scope

It covers: local installation with Docker of SQL Server, RabbitMQ, and Mailpit; the creation of the *LavaRapido* database and its migrations; the start of the six services and the gateway, in containers or from a terminal; the web and mobile applications; the first access, the test accounts, the verification, and the solution of common errors.

It does not cover: setting up the public server on Azure (Deployment Document), continuous integration (GitHub Actions; described in the Deployment Document), building and publishing the APK of the mobile application (Deployment Document), or the functional use of the system, which is explained in the User Manual and the Administrator Manual.

5. Objective

5.1 General Objective

To allow an infrastructure or DevOps person to install and start the Vehicle Washing Management System by following clear, complete, and verifiable instructions.

5.2 Specific Objectives

- Define the hardware, operating system, software, and network requirements to run the complete system.
- Describe how to obtain the code and install the backend, web, and mobile dependencies.
- Explain the configuration of the environment variables and the handling of secrets.
- Detail the creation of the database, the application of the migrations of each service, and the initial data.
- Indicate the startup order and the commands to start, stop, and review the logs of the components.
- Offer verification tests and a guide for solving frequent errors.

6. Minimum Hardware Requirements

The project documentation does not set hardware requirements. The values of Table 2 are recommendations of this manual, calculated to run SQL Server, RabbitMQ, Mailpit, the gateway, and the six services at the same time, plus the development tools. The only minimum with official support is that of SQL Server in a container, which requires at least 2 GB of RAM (Microsoft, n.d.).

Table 2

Recommended Hardware Requirements

Resource	Minimum	Recommended	Remark
Processor	4 cores, 64-bit (x64)	6 or 8 cores	The SQL Server image is for x64; on ARM processors it runs through emulation.
RAM	8 GB	16 GB	With 8 GB, assign at least 5 GB to Docker and close other applications.
Memory for Docker	5 GB	8 GB or more	SQL Server requires 2 GB; each Java service uses up to 75% of the memory of its container.
Free disk	20 GB	40 GB on SSD	Docker images, volumes, Maven, NuGet, and npm dependencies, and logs.
Network	Internet access	Stable broadband	Download of images and dependencies; for a physical phone, the same Wi-Fi network as the computer.

Note. Recommendations of this manual. The minimum of 2 GB of RAM for SQL Server is the one published by Microsoft (n.d.).

6.1 Supported Operating Systems

Table 3

Operating Systems and Additional Requirements

Operating system	Additional requirement
Windows 10 or 11 (64-bit)	Docker Desktop with WSL 2 and virtualization enabled in the BIOS or UEFI.
Ubuntu 22.04 LTS or later	Docker Engine 24 or later with the Docker Compose plugin (v2).
macOS	Docker Desktop; with Apple Silicon, enable x64 emulation (Rosetta) for SQL Server.

Note. Prepared by the authors from the team guide of the infrastructure repository.

6.2 Network Ports

Table 4 lists the ports that the system publishes on the host machine. They must be free before starting the installation.

Table 4

Ports Used by the System

Component	Port	Remark
SQL Server	1433	If a native SQL Server of Windows (service MSSQLSERVER) uses this port, stop it (section 17).
RabbitMQ (AMQP)	5672	Used by the services.
RabbitMQ (management console)	15672	Web interface.
Mailpit (SMTP)	1025	Receives the emails of the services.
Mailpit (web interface)	8025	Shows the captured emails.
API Gateway	8080	Single entry point of the web and mobile applications.
security-service	3001	Accounts, login, and roles.
customer-service	3002	Customers and vehicles.
booking-service	3003	Catalog, schedule, bays, and bookings.
operations-service	3004	Operators, assignment, execution, and ratings.
payment-service	3005	Payment accounts and payments (.NET).
notification-service	3006	Notifications, push, email, and reminders.
Web application (Angular)	4200	Development server.
Mobile application (Expo Metro)	8081	Development server.

Note. Values taken from *docker-compose.yml*, the configuration of each service, and the service catalog.

7. Required Software

Table 5 lists the tools. The versions come from the build files of each repository (*pom.xml*, *Directory.Build.props*, and *package.json*).

Table 5

Software Required for the Installation

Tool	Version	Use
Git	Recent	Obtain the repositories.
Docker Desktop or Docker Engine with Compose v2	24 or later	Run SQL Server, RabbitMQ, Mailpit, the migrations, and, optionally, the services.
JDK (Eclipse Temurin recommended)	21	Run the Java services outside Docker. Maven is not needed: each service includes the Maven Wrapper (*mvnw*).
.NET SDK	8.0	Build and run payment-service.
Node.js and npm	Node.js 20.19 or later (22 LTS recommended); npm 11	Web application (Angular 21) and mobile application.
Angular CLI	21 (optional; *npx ng* works without a global installation)	Development server and build of the web application.
Expo CLI and EAS CLI (through *npx*)	According to *package.json* (Expo SDK 57)	Run the mobile application and build versions with push notifications.
sqlcmd (optional)	Recent	Run verification queries from the host; it is also available inside the SQL Server container.
Graphical SQL client (optional)	SQL Server Management Studio or Azure Data Studio	Visual inspection of the database.
curl	Recent	Verification tests.
Android Studio or an Android phone (optional)	Recent	Emulator or device for the mobile application.

Note. The Liquibase CLI is not needed: the Java services apply their migrations when they start, and the *payment-migrate* container runs those of payment-service.

7.1 Checking the Installed Tools

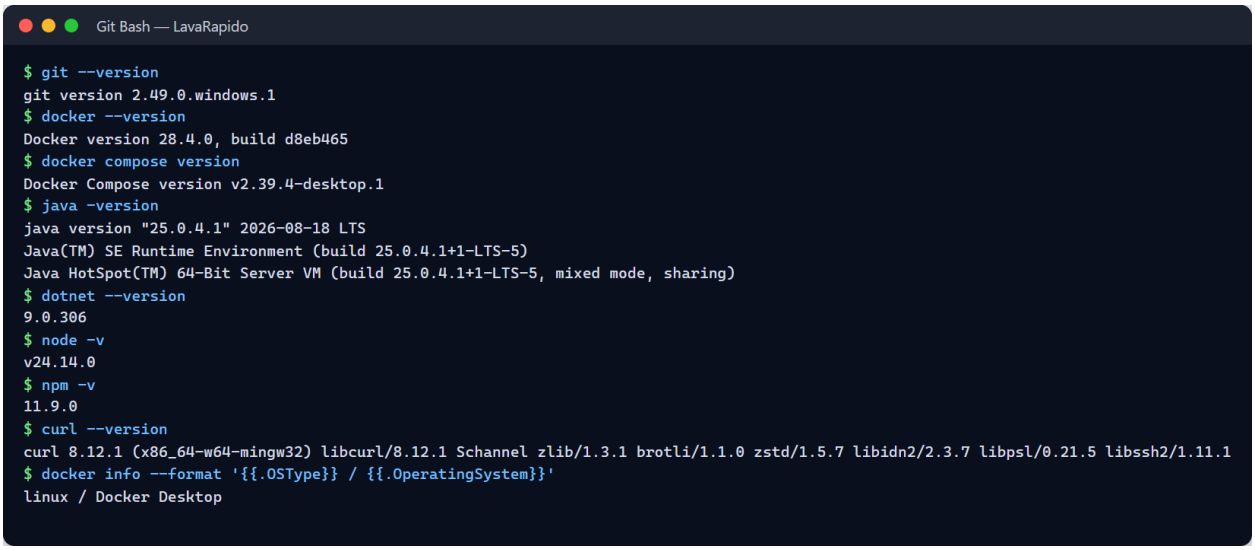
Each command must print a version; if one fails, install the corresponding tool.

```
git --version
docker --version
docker compose version
java -version        # must show 21
dotnet --version     # must show 8.x
node -v
npm -v
curl --version
```

On Windows, *docker info* must show the Linux engine (WSL 2) and not the Windows containers engine.

Figure 1

Verification of Tool Versions



8. Obtaining the Project

The project is distributed in one repository per component. All the backend repositories must be cloned side by side in the same folder and **without changing their names**, because the services look for the shared configuration in *../lavarapido-infra/.env* and *docker-compose.yml* builds them from `../<repository>` (Table 6). Note that the operations repository is called *lavarapido-operation-service* (singular), although the service, its container, and its image are called *operations-service*: if it is cloned with another name, the Compose build fails.

Table 6

Project Repositories

Repository	Content	Address
lavarapido-infra	docker-compose.yml, .env.example, and team guide.	https://github.com/BrayanCABRERA123/lavarapido-infra
lavarapido-security-service	Accounts, login, roles, profile, and password recovery.	https://github.com/BrayanCABRERA123/lavarapido-security-service
lavarapido-customer-service	Customers and vehicles.	https://github.com/camiloguilombo12/lavarapido-customer-service
lavarapido-booking-service	Catalog, schedule, bays, and bookings.	https://github.com/camiloguilombo12/lavarapido-booking-service
lavarapido-operation-service	Operators, assignment, execution, and ratings (service *operations-service*).	https://github.com/BrayanCABRERA123/lavarapido-operation-service
lavarapido-payment-service	Payment accounts, payments, and receipts (.NET 8).	https://github.com/merari0322/lavarapido-payment-service
lavarapido-notification-service	Notifications, push, email, and reminders.	https://github.com/BrayanCABRERA123/lavarapido-notification-service
lavarapido-api-gateway	Gateway (Spring Cloud Gateway).	https://github.com/BrayanCABRERA123/lavarapido-api-gateway
Front-end-proyecto-web	Angular web application.	https://github.com/BrayanCABRERA123/Front-end-proyecto-web
LavadoVehicular-Movil	Android application (React Native with Expo).	https://github.com/merari0322/LavadoVehicular-Movil
vehicle-w-docs	Project documentation, ADRs, and OpenAPI contracts.	https://github.com/code-sena/vehicle-w-docs

Note. The addresses were taken from the Git remotes of the team copies on October 16, 2026.

8.1 Cloning the Repositories

The integration branch of the team is *develop*; the most recent version of every service is there, and *main* holds the delivered versions. On October 23, 2026 *v1.0.0* was tagged in 8 repositories (web, api-gateway, security, customer, booking, operations, notification, and infra) and on

the 24th in payment-service. On October 26, 2026 *v1.1.0* shipped in the 10 repositories, including the mobile app, which had no version until then, and that same day *v1.1.1* in the web, the mobile app, payment-service, and infra. On October 27, 2026 *v1.1.1* shipped in customer-service and booking-service and *v1.1.2* in the web, the mobile app, payment-service, and infra, and that same day payment-service *v1.1.3* was merged into *main*; it was tagged on November 4, 2026. For an evaluation installation, use the most recent tag of each repository: *v1.1.3* in payment-service; *v1.1.2* in the web, the mobile app, and infra; *v1.1.1* in customer-service and booking-service; and *v1.1.0* in the API Gateway, security-service, operations-service, and notification-service. Each one is the tip of *main*, which is protected in the 10 repositories and is what the public server runs (payment-service *v1.1.3* has been in production since November 4, 2026); *develop* has the latest work not yet released in a version.

```
mkdir Backend && cd Backend
git clone -b develop https://github.com/BrayanCABRERA123/lavarapido-infra.git
git clone -b develop https://github.com/BrayanCABRERA123/lavarapido-security-service.git
git clone -b develop https://github.com/camiloguilombo12/lavarapido-customer-service.git
git clone -b develop https://github.com/camiloguilombo12/lavarapido-booking-service.git
git clone -b develop https://github.com/BrayanCABRERA123/lavarapido-operation-service.git
git clone -b develop https://github.com/merari0322/lavarapido-payment-service.git
git clone -b develop https://github.com/BrayanCABRERA123/lavarapido-notification-service.git
git clone -b develop https://github.com/BrayanCABRERA123/lavarapido-api-gateway.git
cd ..
mkdir Frontend && cd Frontend
git clone -b develop https://github.com/BrayanCABRERA123/Front-end-proyecto-web.git
git clone -b develop https://github.com/merari0322/LavadoVehicular-Movil.git
```

To update a copy that already exists, run *git checkout develop* and *git pull* inside each repository.

8.2 Expected Structure

```
Project/
|-- Backend/
|   |-- lavarapido-infra/               (docker-compose.yml + .env.example + .env)
|   |-- lavarapido-security-service/
|   |-- lavarapido-customer-service/
|   |-- lavarapido-booking-service/
|   |-- lavarapido-operation-service/     (singular: name expected by docker-compose.yml)
|   |-- lavarapido-payment-service/
|   |-- lavarapido-notification-service/
|   `-- lavarapido-api-gateway/
`-- Frontend/
    |-- Front-end-proyecto-web/         (Angular)
    `-- LavadoVehicular-Movil/          (React Native / Expo)
```

Figure 2

Repositories Cloned in the Working Folder



9. Installing Dependencies

9.1 Container Images

The local environment uses the images *mcr.microsoft.com/mssql/server:2022-latest*, *rabbitmq:4-management*, *axllent/mailpit*, and *liquibase/liquibase:4.29*; the images of the services are built from their Dockerfiles. From the infrastructure folder, validate the Compose file and

download the base images:

```
cd Backend/lavarapido-infra
docker compose config --quiet && echo "Compose file is valid"
docker compose pull
```

The command *docker compose config* fails if *DB_PASSWORD* or *JWT_SECRET* are missing; in that case, complete the *.env* file first (section 10) and run it again.

9.2 Java Services

The Java services download their dependencies automatically: with Docker, during the build of the image; outside Docker, the first time *mvnw* runs (it can take several minutes). To download them in advance and run the unit tests of a service:

```
cd Backend/lavarapido-security-service
./mvnw -B test            # Windows: .\mvnw.cmd -B test
```

9.3 payment-service (.NET 8)

```
cd Backend/lavarapido-payment-service
dotnet restore
dotnet test               # domain, application, API, and integration tests (xUnit)
```

9.4 Web Application

```
cd Frontend/Front-end-proyecto-web
npm ci
```

The command *npm ci* installs exactly the versions of *package-lock.json*, which makes the installation repeatable. Known vulnerabilities are reviewed with *npm audit*.

9.5 Mobile Application

```
cd Frontend/LavadoVehicular-Movil
npm ci
```

Figure 3

Successful Installation of Dependencies



10. Configuring Environment Variables

The values that change between environments (connection data, secrets, mail, and switches) reside outside the code (Wiggins, 2017). All the services read the same file *lavarapido-infra/.env*: Docker Compose reads it automatically, the Java services import it with *spring.config.import*, and payment-service reads it with its *EnvFile* class. Real environment variables always take precedence over the file. The *.env* file is excluded from Git; the repository only contains *.env.example*.

10.1 Creating the .env File

```
cd Backend/lavarapido-infra
cp .env.example .env          # PowerShell: Copy-Item .env.example .env
git check-ignore -v .env      # must show the .gitignore rule that excludes it
```

If *git check-ignore* prints nothing, the file is not excluded and must not be filled with secrets until the *.gitignore* is corrected.

10.2 Variables of the .env File

Table 7

Main Environment Variables

Variable	Used by	Effect and local value
DB_PASSWORD	SQL Server and all services	Password of the *sa* account. Required; at least 8 characters with uppercase, lowercase, numb symbol, or SQL Server does not start. With *v1.1.3* payment-service does not start without it e
DB_NAME	Infrastructure and services	Database name; *LavaRapido*.
DB_USERNAME	Services	*sa* in the local environment.
DB_URL	Java services run from the IDE	jdbc:sqlserver://localhost:1433;databaseName=LavaRapido;encrypt=true;trustServerCertific
JWT_SECRET	All services	HS256 signing key shared by all services; at least 32 bytes. Required.
CORS_ALLOWED_ORIGINS	Services and gateway	Origins allowed to call from a browser: the web application and Expo web; *http://localhost:4200,http://localhost:8081*.
SPRING_PROFILES_ACTIVE	Java services in Docker	*dev* by default in the Compose file (Swagger and demonstration accounts); use another value as *prod*, in any shared environment. Since *develop* of October 23, 2026, security-service with variable starts in *prod* (E-05).
DEMO_ADMIN_PASSWORD, DEMO_OPERATOR_PASSWORD, DEMO_CLIENT_PASSWORD	security-service (*dev* profile)	Passwords of the demonstration accounts *admin@gmail.com*, *operador@gmail.com*, and *cliente@gmail.com* (since *develop* of October 23, 2026, E-05). Empty: that account is not cre Each person chooses them in their *.env*; they are not in the code or in the documents.
ALLOWED_EMAIL_DOMAINS	security-service	Email domains allowed when signing up, creating users, and changing the email, comma sep *gmail.com* by default (since *develop* of October 23, 2026, A-01). With another domain the se answers 400 *EMAIL_DOMAIN_NOT_ALLOWED*.
ADMIN_EMAIL, ADMIN_PASSWORD, ADMIN_DOCUMENT, ADMIN_PHONE	security-service (production)	First administrator, created only if the database has none; *ADMIN_DOCUMENT* and *ADMIN_PHONE* are their ID number and mobile phone (without them the profile keeps placeholder values). They go in *.env.production* and *docker-compose.prod.yml* passes them.
MAIL_ENABLED	security-service and notification-service	*true* sends emails; *false* writes the recovery code in the log of security-service.
MAIL_HOST, MAIL_PORT, MAIL_USERNAME, MAIL_PASSWORD, MAIL_FROM	security-service and notification-service	SMTP server. Locally Mailpit: localhost, 1025, any user and password. Gmail requires an ap password.
MAIL_DOCKER_HOST	Services in Docker	SMTP host seen from the containers; *mailpit* by default.
MESSAGING_ENABLED	Java services and payment-service	*true* publishes and consumes RabbitMQ events; *false* (default) only writes them in the log. W events, there are no notifications, payment notices, or point notices.
RABBITMQ_HOST, RABBITMQ_PORT, RABBITMQ_USERNAME, RABBITMQ_PASSWORD	Java services and payment-service	localhost, 5672, guest, guest in the local environment. With *v1.1.3*, if *MESSAGING_ENABLED=true* and RabbitMQ is not on *localhost* (Docker or server), payme service does not start without a username and password: *guest* is only accepted with a local RabbitMQ.
PUSH_ENABLED, EXPO_ACCESS_TOKEN	notification-service	*true* sends push notifications through Expo; *false* only writes them in the log.

Variable	Used by	Effect and local value
INTERNAL_API_KEY	security-service, customer-service, booking-service, operations-service, notification-service, and, since October 26, 2026, payment-service	Key of the internal routes; without it, booking, payment, and coupon notices and reminders a sent by email, the notification switches of each user cannot be read, there is no automatic assignment, and operations-service cannot close as No-show the confirmed bookings that en without an operator. Since *develop* of October 23, 2026 (E-01) it is also required for the oper start and finish services: booking-service only accepts those routes from operations-service w key. Since October 24, 2026 customer-service also uses it to ask booking-service whether a v has active bookings before removing it: without it, removing a vehicle answers 503 *BOOKING_SERVICE_UNAVAILABLE*; in production the compose file does not start if it is Since October 26, 2026 payment-service requires it on its internal path (*GET /internal/v1/payments/bookings/{bookingId}/reported*, header *X-Internal-Key*; 401 without a with a wrong one), and *docker-compose.prod.yml* does not start without it: without the key, booking-service cannot check whether an overdue digital booking already has a receipt, and cancels no booking for lack of payment.
REMINDER_LEADS	notification-service	Advance of the reminders; *24h,1h* by default.
BOOKING_ZONE, BOOKING_SLOT_MINUTES, BOOKING_MAX_DAYS_AHEAD	booking-service and operations-service	America/Bogota, 30, and 60 by default.
SESSION_CHECK_ENABLED, SESSION_CHECK_CACHE_TTL, SESSION_CHECK_TIMEOUT	api-gateway	Check of the token session (since October 21, 2026): *true*, *30s*, and *3s* by default. They are commented out in *.env.example* and the Compose file passes them to the container. Since Oct 24 the session is checked with any header scheme.
AUTH_RATE_LIMIT_ENABLED, AUTH_RATE_LIMIT_REQUESTS, AUTH_RATE_LIMIT_WINDOW	api-gateway	Request limit per IP on */api/v1/auth/** (since October 24, 2026): *true*, *60*, and *1m* by default; it, it answers 429 *RATE_LIMITED* with *Retry-After*. They have a default value in the service compose file does not pass them: to change them, add them to the gateway's *environment* blo
TRUSTED_PROXIES	api-gateway	Networks from which the *X-Forwarded-For* header is trusted (by default the private and loca *127.0.0.0/8*, *10.0.0.0/8*, *172.16.0.0/12*, *192.168.0.0/16*, *::1*, and *fc00::/7*). With Caddy in fron does not need to be changed.
ROUTE_CONNECT_TIMEOUT, ROUTE_READ_TIMEOUT	api-gateway	Time to connect and to read the answer of each service (since October 24, 2026): *5s* and *60s* default; if exceeded, the gateway answers 500.
BOOKING_EXPIRY_CHECK_ENABLED, BOOKING_EXPIRY_CHECK_INTERVAL	booking-service	**No effect since October 24, 2026** (B-07): the task that cancelled expired scheduled bookings removed, because every booking is created and rescheduled as confirmed. The Compose file passes them; they can be deleted from *.env*.
BOOKING_EXPIRY_GRACE_MINUTES	booking-service and operations-service	Minutes after the end time before an expired booking is closed; *60* by default. Both services so if it is changed it must have the same value in both; the Compose file passes it to both.
FOLLOW_UP_ENABLED, FOLLOW_UP_INTERVAL, FOLLOW_UP_LOOK_BACK_DAYS	operations-service	Review of the bookings that ended without being closed (since the night of October 21, 2026 confirmed booking that never had an operator becomes No-show, and one that had an operat nobody recorded stays in "To close". *true*, *PT5M*, and *28* (days it looks back, because the ser stay off for weeks) by default; needs *INTERNAL_API_KEY*. They go through the Compose f
START_LATE_LIMIT	operations-service	How long after the end time the operator can start a service (since October 24, 2026): *PT4H* default; afterwards it answers 409 *SERVICE_TOO_LATE*. It does not go through the compos
BOOKING_CASH_ACTIVE_LIMIT	booking-service	Maximum active cash bookings per customer (since October 26, 2026); *1* by default; beyond *CASH_BOOKING_LIMIT*. Commented out in *.env.example*; the compose file passes it to the container.
BOOKING_PAYMENT_WINDOW	booking-service	Deadline to upload the receipt of a booking with digital payment (or the start time, if that cor first); *30m* by default. To test the expiry locally without waiting, it can be lowered (for exam *1m*).
BOOKING_PAYMENT_EXPIRY_ENABLED, BOOKING_PAYMENT_EXPIRY_INTERVAL	booking-service	*PaymentDeadlineJob*, which cancels overdue digital bookings with *PAYMENT_TIMEOUT* a asking payment-service: *true* and *60s* by default. They go through the compose file.
SECURITY_SERVICE_URL, CUSTOMER_SERVICE_URL, BOOKING_SERVICE_URL, PAYMENT_SERVICE_URL	Services that call others	Addresses of the services; they have defaults for the local environment. payment-service nee BOOKING_SERVICE_URL to read the total and to credit the points of a booking. Since Oct 24, 2026 customer-service also uses it, but with the service root (*http://booking-service:3003* Docker, *http://localhost:3003* locally, without */api/v1*), fixed in the compose file; optional, *BOOKING_SERVICE_TIMEOUT* (3 s). Since October 26, 2026 booking-service uses *PAYMENT_SERVICE_URL* (*http://payment-service:3005* in Docker, fixed in the compose fil *http://localhost:3005* by default locally) for the internal query to payment-service.

Note. Prepared by the authors from *.env.example*, *docker-compose.yml*, and the configuration files of the services.

10.3 Generating Secrets

The JWT secret, the *sa* password, and the internal key must be generated with a secure random source, not typed by hand:

```
openssl rand -base64 48                                  # JWT_SECRET
node -e "console.log(require('crypto').randomBytes(32).toString('base64url'))"   # INTERNAL_API_KEY
```

The generated value is pasted in the *.env* file, never in a message, a shared document, or a repository.

10.4 Example of the .env File

The following fragment shows the structure; secrets are represented with asterisks.

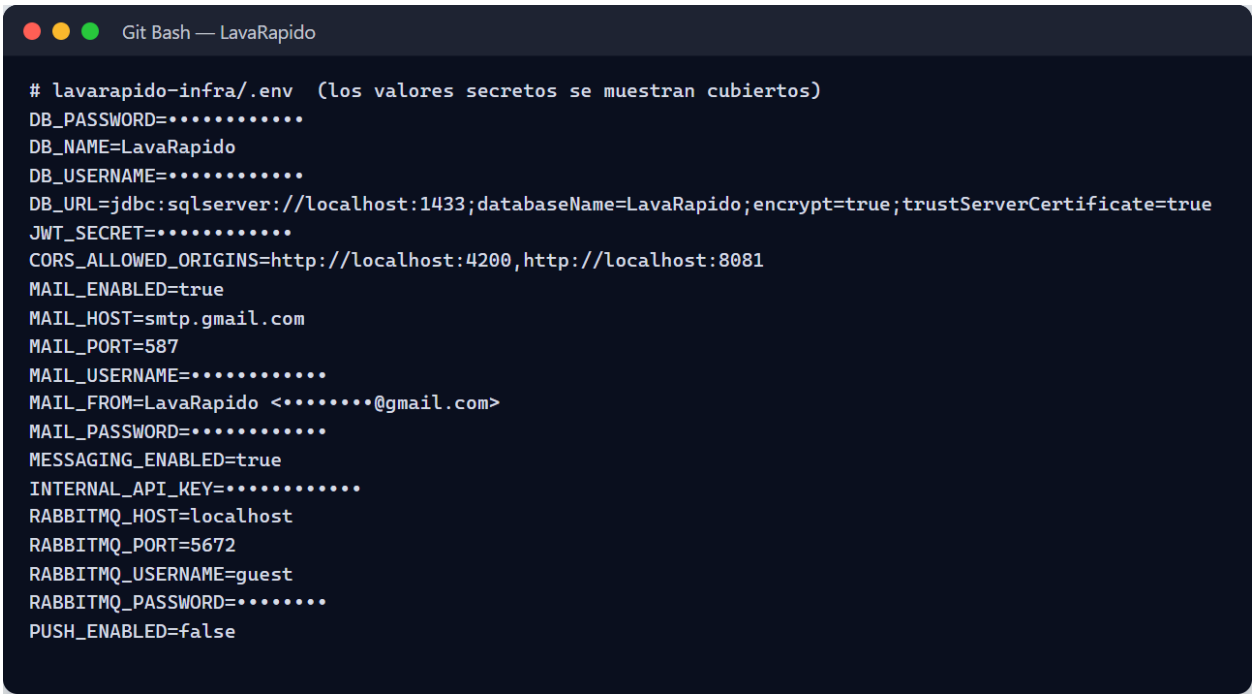
```
DB_PASSWORD=********
DB_NAME=LavaRapido
DB_USERNAME=sa
DB_URL=jdbc:sqlserver://localhost:1433;databaseName=LavaRapido;encrypt=true;trustServerCertificate=true
JWT_SECRET=********
CORS_ALLOWED_ORIGINS=http://localhost:4200,http://localhost:8081
MAIL_ENABLED=true
MAIL_HOST=localhost
MAIL_PORT=1025
MAIL_USERNAME=dev
MAIL_PASSWORD=dev
MAIL_FROM=LavaRapido <no-reply@lavarapido.local>
MESSAGING_ENABLED=true
RABBITMQ_HOST=localhost
RABBITMQ_PORT=5672
RABBITMQ_USERNAME=guest
RABBITMQ_PASSWORD=guest
PUSH_ENABLED=false
INTERNAL_API_KEY=********
```

10.5 Configuring the Client Applications

- **Web application.** It calls the relative address */api/v1*. In development, *proxy.conf.json* forwards every request that starts with */api* to the gateway (*http://localhost:8080*), so no file has to be edited.
- **Mobile application.** Copy *.env.example* to *.env.local* and set *EXPO_PUBLIC_API_URL*. In the Android emulator use *http://10.0.2.2:8080/api/v1* (the default value); on a physical phone, the IPv4 address of the computer on the same Wi-Fi network (for example, *http://192.168.1.10:8080/api/v1*); and for the web version of Expo, *http://localhost:8080/api/v1*. After changing it, restart Expo with *npx expo start --clear*. The optional *EXPO_PUBLIC_WEB_URL* variable sets the web address the app uses to build the public link of the inspection report; when it is not set, the same host as the gateway with port 4200 is used.

Figure 4

Configured .env File With Hidden Secrets



Note. Do not capture real values of secrets.

11. Database Creation and Import

The system uses a single SQL Server 2022 database called *LavaRapido*, divided into nine schemas: *security*, *audit*, *customer*, *catalog*, *booking*, *execution*, *promotion*, *payment*, and *notification* (58 tables). Each service owns its schemas and migrates only them with Liquibase, with its own control tables `DATABASECHANGELOG_<SERVICE>` (Vehicle Washing Development Team, 2026a).

11.1 Starting SQL Server and Creating the Database

```
cd Backend/lavarapido-infra
docker compose up -d
docker compose ps
```

This command starts, without the services:

- *sqlserver* (container *lavarapido-sqlserver*), with a health check and the volume *sqlserver-data*.
- *db-init*, which runs once and creates the *LavaRapido* database if it does not exist.
- *payment-migrate*, which runs once the Liquibase changelogs of payment-service (schemas *payment* and *promotion*).
- *mailpit* and *rabbitmq* (container *lavarapido-rabbitmq*, with a health check).

Wait until *lavarapido-sqlserver* and *lavarapido-rabbitmq* show the status *healthy* and *db-init* and *payment-migrate* end with exit code 0. If the SQL Server container stops after a few seconds, the usual causes are a weak *DB_PASSWORD*, insufficient memory, or port 1433 already in use (section 17).

Figure 5

Infrastructure Containers Running



11.2 Migrations of Each Service

The Java services apply their migrations automatically when they start (*spring.liquibase*), and then Hibernate only validates that the entities match the tables. Therefore, the tables of the schemas *security*, *audit*, *customer*, *catalog*, *booking*, *execution*, and *notification* are created the first time each service starts (section 12). Table 8 shows the changelogs of the version of October 21, 2026.

Table 8

Changelogs of Each Service

Service	Changelogs	Control table
security-service	001 schemas, 002 security and audit core, 003 user preference, 020 theme palettes, 023 custom role (created and dropped again by 026), 026 role permissions (permission catalog and default assignments), 030 notification switches of the user preference, 034 lockout after failed attempts, 035 person address, 036 account creation date, 039 acceptance of the legal documents; seed 101 roles.	DATABASECHANGELOG_SECURITY
customer-service	004 customer schema, 014 customer user id, 015 vehicle model; seed 102 vehicle types.	DATABASECHANGELOG_CUSTOMER
booking-service	005 catalog and booking schemas, 018 changes of ADR-010, 021 schedule history, 022 loyalty points of the catalog, 024 removal of the bay operator, 032 business data of the establishment; seeds 103 catalog, 104 statuses and reasons, 109 establishment, schedule, and four bays, 110 business data, 111 removal of the example business data so the real business fills it in, and 112 *EXPIRED* reason of the automatic close of expired bookings; since October 24, 2026, 044 price greater than 0, 045 optimistic locking of the booking, and seed 113 removal of the sample data of the site; since October 26, 2026, 049 payment mode and deadline of the booking (*payment_mode*, *payment_deadline*, *payment_reported_at*, and index *ix_booking_payment_pending*) and seed 117 *PAYMENT_TIMEOUT* reason.	DATABASECHANGELOG_BOOKING
operations-service	007 execution schema, 033 inspection reports with phases, findings, and photos, 037 operator profile, 040 time of the service-not-recorded notice (*follow_up_notice_at*); seed 105 execution statuses.	DATABASECHANGELOG_OPERATIONS
notification-service	010 notification schema, 016 device tokens, 017 processed events, 019 booking reminders, 025 promotion redeemed type, 027 staff notifications and booking tracking, 028 payment refunded type, 029 loyalty and inspection types, 031 unassigned booking alert, 038 new app version notice type, 041 service-not-recorded notice types for the operator and the administrator; seed 108 notification types.	DATABASECHANGELOG_NOTIFICATION
payment-service	db/changelog/changes/001 to 020 (own numbering; 017 package fields, 018 required points, 019 catalog verification, 020 points reversal and sequence), run by the *payment-migrate* container.	DATABASECHANGELOG_PAYMENT

Note. Source: migration strategy of the project documentation and the changelogs of the repositories.

Rules that must always be respected. Changesets are forward-only. A changeset that already ran is never edited: Liquibase calculates a checksum and refuses to continue if the content changed, which is an intentional protection. Every changeset has a precondition that marks it as run if the object already exists. The initial data uses fixed identifiers, and two of them are critical: *booking_status_id = 5* (CANCELLED) is used by the constraint *ck_booking_cancel* and the trigger *tr_booking_bay_overlap*, and *payment_status_id = 3* (APPROVED) by the index *ux_payment_one_approved_per_booking*.

11.3 Alternative Method: Global Script

As of October 23, 2026 the script *lavarapido-6-services-sqlserver.sql* is not versioned in any of the repositories (not in their history either), so this method cannot be followed: the database is created only with the Liquibase changelogs each service applies on startup (section 11.2). If the team publishes the script again, it is run on the empty database and, because every changeset has a precondition, Liquibase marks as run the objects that already exist:

```
sqlcmd -S localhost,1433 -U sa -P "<SA_PASSWORD>" -C -d LavaRapido -b -i lavarapido-6-services-sqlserver.sql
```

The recommended method is the one of section 11.2, because it leaves the history of each service in its control table.

11.4 Database Accounts

In the local environment all the services connect with the *sa* account (variable *DB_USERNAME*). The security policy of the project requires that each service use a login with rights only on its own schemas; it is a pending improvement that must be applied before any shared deployment. The generic procedure of SQL Server is:

```
CREATE LOGIN [booking_svc] WITH PASSWORD = '<STRONG_PASSWORD>', CHECK_POLICY = ON;
USE [LavaRapido];
CREATE USER [booking_svc] FOR LOGIN [booking_svc];
GRANT SELECT, INSERT, UPDATE, DELETE, EXECUTE ON SCHEMA::[catalog] TO [booking_svc];
GRANT SELECT, INSERT, UPDATE, DELETE, EXECUTE ON SCHEMA::[booking] TO [booking_svc];
```

Note. The migrations require rights to create objects (DDL); if the services stop using *sa*, the migrations must be run with a separate account.

11.5 Verifying the Structure and the Initial Data

The queries can be run from the host with *sqlcmd* or from inside the container:

```
docker compose exec sqlserver /opt/mssql-tools18/bin/sqlcmd -S localhost -U sa -P "$DB_PASSWORD" -C -d LavaRapido -Q "<QUERY>"
```

```
-- 1) Tables per schema (58 in total when all the services have started)
SELECT s.name AS schema_name, COUNT(*) AS tables
FROM sys.tables t JOIN sys.schemas s ON s.schema_id = t.schema_id
WHERE s.name IN ('security','audit','customer','catalog','booking',
                 'execution','promotion','payment','notification')
GROUP BY s.name ORDER BY s.name;

-- 2) Migration history of each service
SELECT name FROM sys.tables WHERE name LIKE 'DATABASECHANGELOG[_]%';

-- 3) Critical fixed identifiers
SELECT booking_status_id, code FROM booking.booking_status WHERE booking_status_id = 5;
SELECT payment_status_id, code FROM payment.payment_status WHERE payment_status_id = 3;
```

Table 9

Expected Results of the Initial Data

Table	Expected result
security.role	3 rows: ADMIN, OPERATOR, and CLIENT.
customer.vehicle_type	6 rows: CAR, SEDAN, SUV, PICKUP, TRUCK, and MOTO.
catalog.service	The services BASIC, PREMIUM, and FULL with prices for the six vehicle types.
booking.booking_status	6 rows; identifier 5 corresponds to CANCELLED.
booking.cancellation_reason	7 rows; *EXPIRED* (seed 112) is only used by the system and does not appear in the list of reasons.
booking.service_bay	4 bays.
execution.execution_status	5 rows: PENDING, IN_PROGRESS, PAUSED, COMPLETED, and WITH_ISSUE.
payment.payment_status	5 rows; identifier 3 corresponds to APPROVED.
payment.payment_method_type	4 rows: EFECTIVO, NEQUI, DAVIPLATA, and TRANSFERENCIA.
payment.loyalty_movement_type	5 rows: EARNED, REDEEMED, EXPIRED, ADJUSTED, and REVERSED.
notification.notification_type	26 rows.
security.permission	The permission catalog loaded by changelog 026.

Note. If a catalog table is empty, the service that owns it has not started yet or its seed changeset did not run.

11.6 Backup and Reset of the Local Database

Before applying changes in an environment with data that matters, back up the database (Administrator Manual, section 14):

```
docker compose exec sqlserver /opt/mssql-tools18/bin/sqlcmd -S localhost -U sa -P "$DB_PASSWORD" -C \
  -Q "BACKUP DATABASE [LavaRapido] TO DISK = N'/var/opt/mssql/data/backup.bak' WITH INIT, CHECKSUM"
docker cp lavarapido-sqlserver:/var/opt/mssql/data/backup.bak ./backup.bak
```

On the public server backups are automatic: *deploy/backup-db.sh* backs up the database every day and before every update, *deploy/verify-backup.sh* checks each backup by restoring it under another name, and *deploy/restore-db.sh* restores one (Deployment Document, section 3.2).

To reset a disposable local environment, stop the containers and delete the volumes. **This deletes all the data:**

```
docker compose down -v
```

12. Running the Backend

12.1 Startup Order

The services do not depend on a strict order beyond the infrastructure, but the sequence of Table 10 avoids errors in the first start (for example, a booking that cannot validate the vehicle because customer-service is not ready).

Table 10

Startup Sequence of the Local Environment

Step	Component	Ready when
1	SQL Server, RabbitMQ, and Mailpit (*docker compose up -d*)	sqlserver and rabbitmq are *healthy*; db-init and payment-migrate finished.
2	security-service (3001)	The log shows "Started SecurityServiceApplication".
3	customer-service (3002)	It responds on /actuator/health.
4	booking-service (3003)	It responds on /actuator/health.
5	operations-service (3004)	It responds on /actuator/health.
6	payment-service (3005)	It responds on /actuator/health.
7	notification-service (3006)	It responds on /actuator/health; it creates its queue if MESSAGING_ENABLED is true.
8	api-gateway (8080)	It responds on /actuator/health.

Note. The first start of each Java service takes longer because it downloads dependencies and creates its tables.

12.2 Option A: Services in Docker (Recommended)

With the profile *app*, Docker Compose builds and starts the gateway and the six services (the five Java services and payment-service) in containers, after the infrastructure:

```
cd Backend/lavarapido-infra
docker compose --profile app up -d --build
docker compose ps
```

The *--build* option rebuilds the images from the current code; omit it to reuse images already built. Since October 8, 2026, payment-service has its own Dockerfile (build with the .NET 8 SDK, execution with the ASP.NET 8 runtime and an unprivileged user) and starts after *payment-migrate* ends successfully; the *api-gateway* container receives *ROUTE_PAYMENT_URL: http://payment-service:3005*, so the limitation TD-014 of version 3.0 no longer exists.

Figure 6

Environment Containers Running



12.3 Option B: Java Services From a Terminal or the IDE

For development or debugging, keep only the infrastructure in Docker and run each service with the Maven Wrapper, one terminal per service. The services read *../lavarapido-infra/.env* by themselves and use the *dev* profile by default, which enables Swagger and the demonstration accounts.

```
cd Backend/lavarapido-security-service
./mvnw spring-boot:run          # Windows: .\mvnw.cmd spring-boot:run
```

Repeat the command in *lavarapido-customer-service*, *lavarapido-booking-service*, *lavarapido-operation-service*, *lavarapido-notification-service*, and *lavarapido-api-gateway*. Do not run the same service in Docker and on the host at the same time, because both would try to use the same port.

12.4 payment-service (.NET 8) Outside Docker

```
cd Backend/lavarapido-payment-service
dotnet run --project src/PaymentService.Api       # http://localhost:3005
```

This option is for development or debugging; with Option A the service already runs in its container (stop it first with *docker compose stop payment-service*). The service reads *lavarapido-infra/.env*, listens on *PAYMENT_PORT* (3005 by default), and does not start without *JWT_SECRET* (since *v1.1.3*, not without *DB_PASSWORD* either). It never creates its tables: they are created by the *payment-migrate* container (section 11.1). The repository also has its own *docker-compose.yml* (since *v1.1.3*, SQL Server 2022 on port 1434 with the database *LavaRapidoPaymentIT*, the same as the CI "Pruebas de integración" job; its password is *MSSQL_SA_PASSWORD* in that repository's *.env*) only to test the migrations and the integration tests; it must not be used to run the system.

12.5 RabbitMQ and Mail

With *MESSAGING_ENABLED=true*, the services create the RabbitMQ topology when they start (payment-service only publishes: payment results, redeemed coupons, and points earned): the topic exchange *carwash.events*; the queues *customer-service.events*, *notification-service.events*, and *operations-service.booking-confirmed* (only *booking.confirmed*; it replaced the old *operations-service.events*, which operations-service empties and deletes at start-up); and their dead-letter queues (*.dlq*, through the *carwash.events.dlx* exchange), to which a message moves after three failed retries. Since October 26, 2026 booking-service also consumes: it creates the *booking.payment-events* queue (keys *payment.reported* and *payment.confirmed*) and its *booking.payment-events.dlq*. If payment-service runs outside Docker on Windows and does not connect to RabbitMQ, use *RABBITMQ_HOST=127.0.0.1* instead of *localhost*: .NET may resolve *localhost* over IPv6 and Docker Desktop cuts the AMQP connection (finding of the test of October 26, 2026). With *MESSAGING_ENABLED=false*, the system works, but customer profiles are created on the first request of the customer and the notifications that depend on events are not generated.

With *MAIL_ENABLED=true* and Mailpit, the emails (welcome, recovery code, booking, payment, and coupon notices, inspection reports, and reminders) do not reach real inboxes: they are read at *http://localhost:8025*. With *MAIL_ENABLED=false*, the recovery code is written in the log of security-service.

12.6 Logs, Stopping, and Updating

```
docker compose logs -f security-service          # follow a service's log
docker compose logs --since 10m booking-service  # last 10 minutes
docker compose stop                              # stop without deleting data
docker compose start                             # resume
docker compose down                              # remove containers, keep volumes
docker compose --profile app up -d --build <service>   # update a service after a change
```

The logs include the time, the level, the service name, and the correlation identifier. A log must never contain passwords, tokens, or receipt content; if they appear, it is a defect that must be reported.

13. Running the Frontend

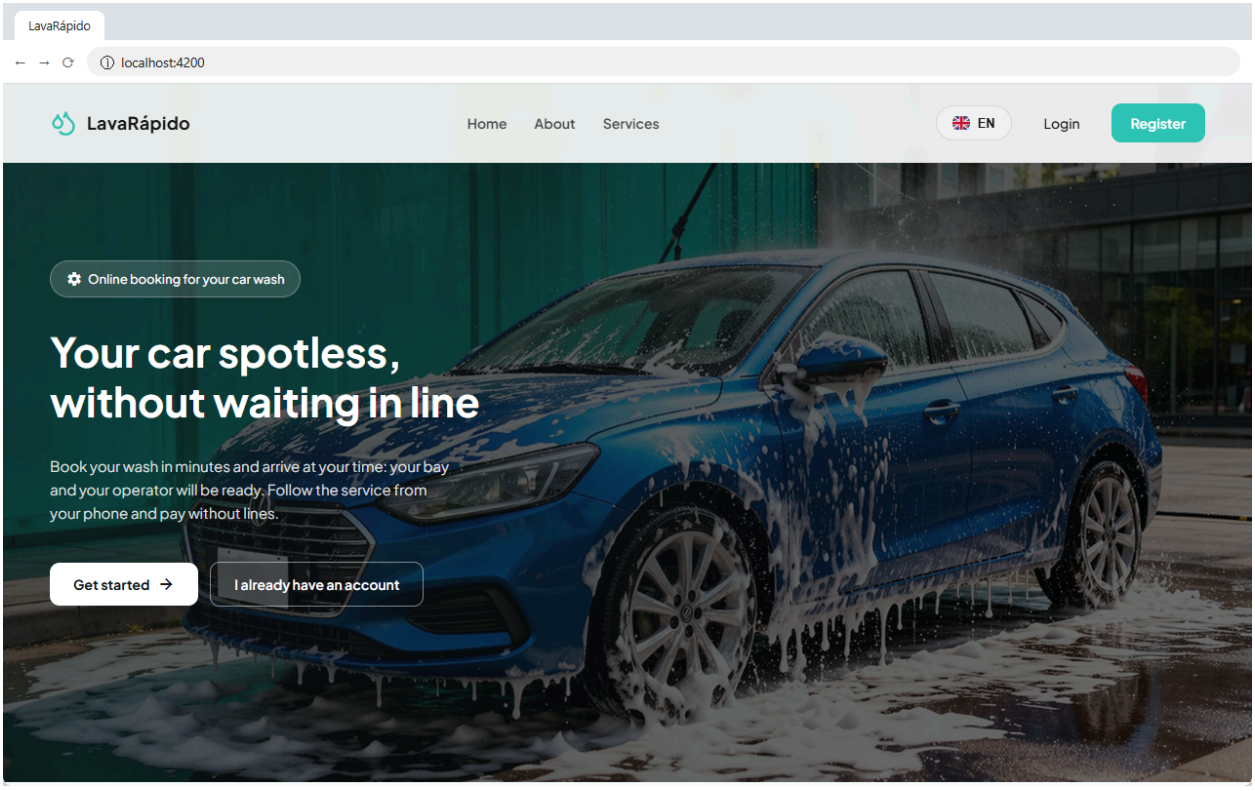
13.1 Web Application (Angular)

```
cd Frontend/Front-end-proyecto-web
npm start                    # equivalent to: npx ng serve
```

The development server publishes the application at *http://localhost:4200* and forwards */api* to the gateway. The production build is generated with *npm run build* and leaves the static files in *dist/*; publishing them on a web server is outside the scope of this manual.

Figure 7

Web Application Running



13.2 Mobile Application (React Native With Expo)

```
cd Frontend/LavadoVehicular-Movil
npx expo start               # or: npm start
```

The command opens the Metro development server (port 8081) and shows a QR code. For a quick test, open the application in an Android emulator (key *a*) or on a phone with Expo Go on the same Wi-Fi network. Remember to configure *EXPO_PUBLIC_API_URL* (section 10.5).

New native dependencies. Since October 17, 2026, the app uses *expo-image-picker* (camera and gallery) and *expo-image-manipulator* (shrinking the photos) for the inspection report. After *git pull*, run *npm install* in *Frontend/LavadoVehicular-Movil*. Expo Go already includes these modules, but an APK built before that date does not: a new version must be built and published with EAS, as explained in the Deployment Document (section 4.3.3, "Publishing the mobile app"). The camera and photo permissions are declared in the *expo-image-picker* plugin inside *app.json*. The changes of October 21, 2026 (compressed receipt, QR magnifier, and mandatory APK update screen) add no native dependencies.

Push notifications. Expo Go does not receive remote push notifications. To test them, build the APK with the EAS *preview* profile (the only one defined in *eas.json* besides *production*; there is no *development* profile) and set *PUSH_ENABLED=true* in the backend:

```
npx eas-cli login
npx eas-cli build --profile preview --platform android
```

Figure 8

Expo Development Server



13.3 Static Mockups (Optional)

The documentation repository includes HTML mockups of the screens in *12-ux-ui/mockup*. They are opened directly in the browser and do not require the backend; they must not be confused with the real application.

14. Accessing the System

Once all the components are started, people access the system through the web or mobile application, and the technical staff also uses the support interfaces of Table 11.

Table 11

Access Points of the Local Environment

Access	Address	Who uses it
Web application	http://localhost:4200	Customers, operators, and administrators.
Mobile application	Expo Go or development build	Customers, operators, and administrators.
API Gateway	http://localhost:8080/api/v1	Client applications and API tests.
Swagger of the Java services (profile dev)	http://localhost:3001/swagger-ui.html (3002, 3003, 3004, and 3006 equivalent)	Developers.
Swagger of payment-service (Development environment)	http://localhost:3005/swagger	Developers.
Mailpit	http://localhost:8025	Technical staff: recovery codes and other emails.
RabbitMQ console	http://localhost:15672	Technical staff: exchange, queues, and failed messages.
SQL Server	localhost,1433	Technical staff: sqlcmd or a graphical client.

Note. Values taken from the configuration of the services and the team guide.

14.1 First Access to the Web Application

1. Open an updated browser at *http://localhost:4200*.
2. Check that the welcome screen shows the **Login** and **Register** buttons.
3. Click **Login** (route */auth*) and log in with a test account (section 15).
4. Check that the system opens the home screen of the role: */client*, */operator*, or */admin*.

15. Test Users

15.1 Demonstration Accounts

When security-service starts with the *dev* profile (the default one in local), it creates one demonstration account per role (administrator *admin@gmail.com*, operator *operador@gmail.com*, and customer *cliente@gmail.com*) if they do not exist. Since *develop* of October 23, 2026 (E-05) their passwords are not in the code or in any document: each person defines them in their *lavarapido-infra/.env* with *DEMO_ADMIN_PASSWORD*, *DEMO_OPERATOR_PASSWORD*, and *DEMO_CLIENT_PASSWORD*, and the account whose variable is empty is not created (team guide, *lavarapido-infra/GUIA-EQUIPO.md*, section 5). Accounts already created with the old passwords keep working until they are changed. Those accounts must never exist in a shared environment.

15.2 Rules for Accounts

Table 12

Rules for the Data of an Account

Data	Rule
Email	Must end in @gmail.com; no spaces. Each email admits a single account.
ID number	Between 5 and 20 digits; unique.
First and last names	Minimum 3 and 2 characters, respectively.
Phone	Colombian mobile number of 10 digits starting with 3.
Password	8 to 72 characters, with at least one uppercase letter, one number, and one special character, without spaces.
Session	The token lasts one hour; after that, the person logs in again.
Failed attempts	Five failed attempts within 15 minutes lock the account for 15 minutes; resetting the password unlocks it.

Note. Source: validations of the registration form and the password policy of security-service.

15.3 Creating a Test Customer

1. Open the web application and click **Register**.

2. Fill in the form with test data and a **test** Gmail email; accept the legal documents.
3. Click **Register** and then **Go to login**.
4. With *MAIL_ENABLED=true* and *MESSAGING_ENABLED=true*, open Mailpit (*http://localhost:8025*) and check that the welcome email arrived.
5. Log in and check that the customer dashboard (*/client*) opens.

15.4 Operator and Administrator Accounts

Operator and administrator accounts are created by an administrator from **Management > Users > Create user** or **Operators > New operator**, with a temporary password (Administrator Manual, section 9.3). The first administrator of an environment without the *dev* profile is created with the procedure of section 8.4 of the Administrator Manual.

16. Installation Verification

The verification is done in layers: infrastructure, database, messaging, services, and functional flow (Table 13).

Table 13

Installation Checklist

No.	Check	Expected result
1	Containers	sqlserver and rabbitmq *healthy*; mailpit and the services running.
2	Database	The 58 tables in the nine schemas (section 11.5).
3	Initial data	The results of Table 9.
4	RabbitMQ	Exchange *carwash.events* and the queues of customer and notification (with messaging active).
5	Mailpit	Web interface available.
6	Health of each service	/actuator/health responds {"status":"UP"} in ports 3001 to 3006 and 8080.
7	Protected route without a token	Error 401 returned by the service.
8	Web application	The welcome screen and the login are displayed.
9	Registration and login	An account is created and the customer dashboard opens.
10	Password recovery	The code arrives in Mailpit and allows changing the password.
11	Booking flow	A vehicle is registered and a booking is created.
12	Assignment and execution	The administrator assigns an operator, who starts and finishes the service.
13	Payment	The customer reports a payment and the administrator approves it.
14	Persistence	The data remains after restarting the containers.

Note. Prepared by the authors from the integration checks of the project documentation.

16.1 Infrastructure and Database

```
docker compose ps
docker compose exec sqlserver /opt/mssql-tools18/bin/sqlcmd -S localhost -U sa -P "$DB_PASSWORD" -C -Q "SELECT @@VERSION"
```

Then run the queries of section 11.5. If the results do not match, do not continue with the following tests.

16.2 Messaging and Mail

```
curl -s -u guest:guest http://localhost:15672/api/exchanges/%2F/carwash.events
docker compose exec rabbitmq rabbitmqctl list_queues name messages
curl -s -o /dev/null -w "%{http_code}\n" http://localhost:8025/
```

The first command returns the data of the exchange (type *topic*); the second, the queues with their messages (the *.dlq* queues must be at zero); and the third, the code 200 of Mailpit. Replace *guest:guest* with the RabbitMQ credentials of the *.env* file if they were changed.

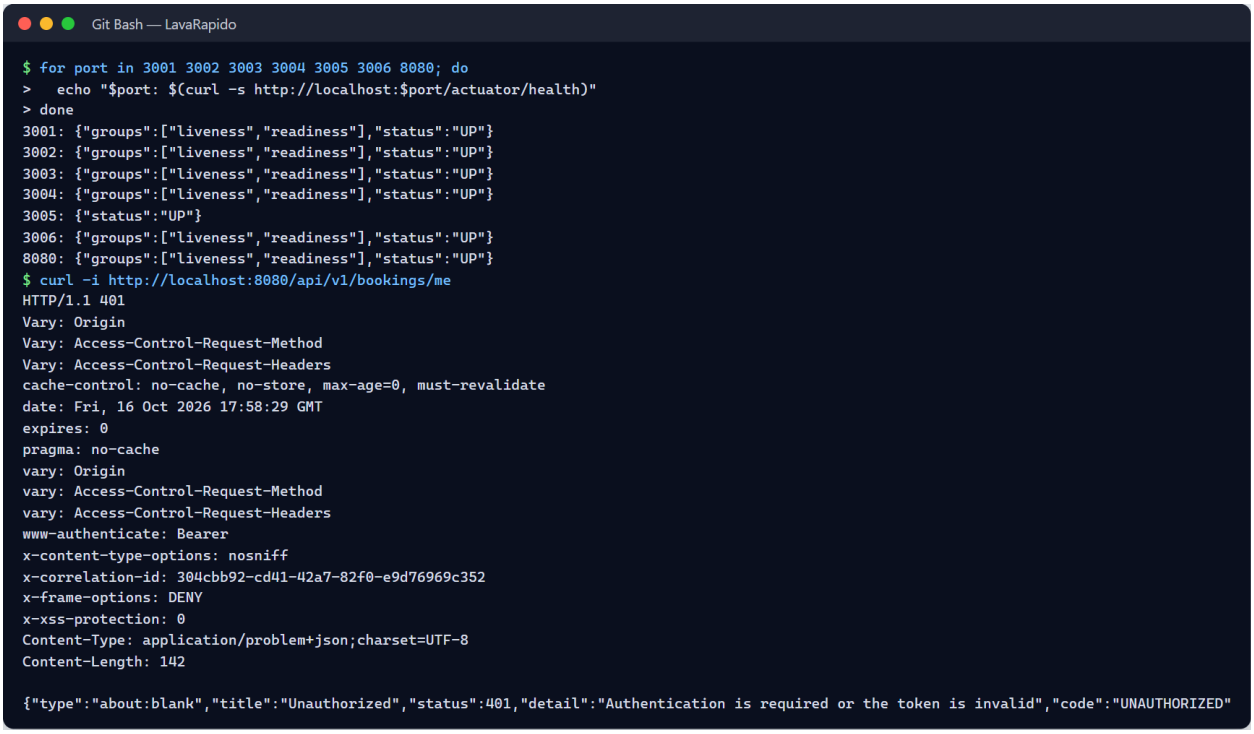
16.3 Services and Gateway

```
for port in 3001 3002 3003 3004 3005 3006 8080; do
  echo "$port: $(curl -s http://localhost:$port/actuator/health)"
done
curl -i http://localhost:8080/api/v1/bookings/me
```

Each health endpoint must answer *{"status":"UP"}*. The last command calls a protected route without a token through the gateway: it must return **401**. The gateway does not check the token signature (ADR-006): a request without a token goes straight through and booking-service rejects it. Since October 21, 2026 the gateway does cut the token of a closed session: after signing out, repeating the call with the same token (*-H "Authorization: Bearer `<TOKEN>`"*) returns 401 from the gateway, without reaching the service. Every service, including payment-service, checks the database in its health endpoint: if SQL Server does not answer, it replies *DOWN* with code 503.

Figure 9

Response of the Health Endpoints



16.4 Minimum Functional Flow

1. Register a test customer and check the welcome email in Mailpit.
2. Test password recovery: request the code, read it in Mailpit, and change the password.
3. Log in as the customer, register a vehicle, and create a booking; a free slot is confirmed immediately.
4. Log in as the administrator and assign an operator to the booking (**Bookings > Assign**).
5. Log in as the operator on the day of the booking, start the service, and finish it.
6. As the customer, report a payment with a test receipt; as the administrator, approve it in **Payments**.
7. With messaging active, check that the customer received the notifications of the booking, the assignment, and the approved payment, the email in Mailpit, and the points in the dashboard.
8. Sign out and repeat a call with the same token: the gateway answers 401.
9. Optional: as administrator, create a test booking for a time that ends soon, with an operator, and do not start it. A few minutes after its end time it must appear in the **To close** card of the administrator's home, with the notice "Servicio sin registrar" (service not recorded); clicking **Attended** makes it Completed.
10. Restart the containers (*docker compose restart*) and check that the data remains.

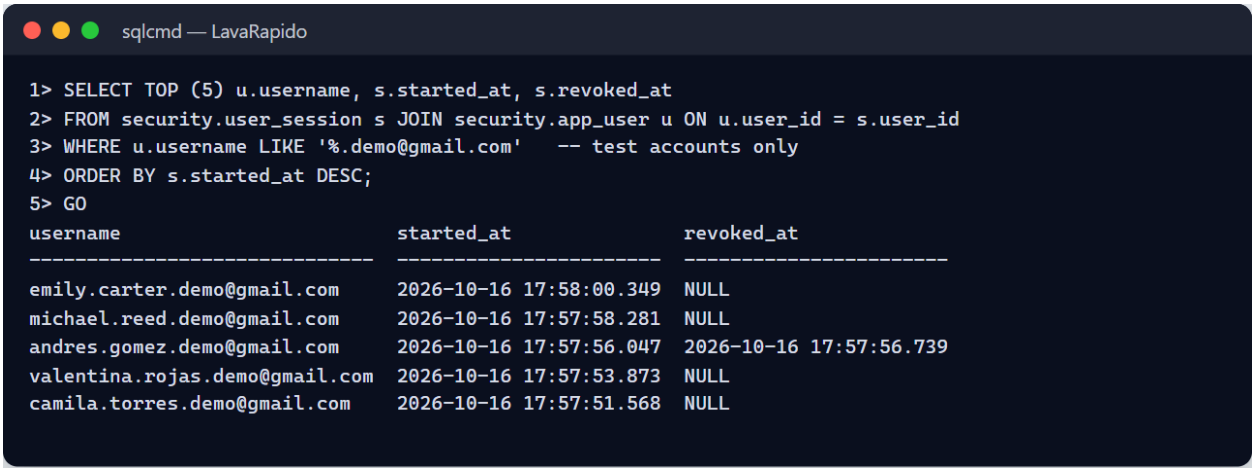
16.5 Traces Verification

The audit table *audit.audit_log* exists but is not populated in the current version (Administrator Manual, section 16). After the tests of section 16.4, verify instead that the sessions were recorded:

```
SELECT TOP (5) u.username, s.started_at, s.revoked_at
FROM security.user_session s JOIN security.app_user u ON u.user_id = s.user_id
ORDER BY s.started_at DESC;
```

Figure 10

Session Records



17. Common Errors and Solutions

The first step in any error is to consult the log of the affected service with *docker compose logs --tail 100 <service>* or the terminal where it runs (Table 14).

Table 14

Frequent Errors, Probable Causes, and Solutions

Error or symptom	Probable cause	Solution
Login failed for user 'sa'	A native SQL Server of Windows answers on port 1433, or the password differs from the one used when the volume was created.	Stop the Windows service MSSQLSERVER; use the original password or recreate the volume (*docker compose down -v*) in a disposable environment.
A port is in use (address already in use)	Another process uses the port.	Find it (*netstat -ano* on Windows, *lsof -i :PORT* on Linux or macOS) and stop it.
The SQL Server container stops	Weak DB_PASSWORD or less than 2 GB of memory.	Use a strong password and assign more memory to Docker.
docker compose config fails	DB_PASSWORD or JWT_SECRET missing in .env.	Complete the .env file (section 10).
A Java service does not start: "JWT secret" or database errors	The .env file is not in *../lavarapido-infra/* or the folder was renamed.	Keep the repositories side by side with their original names.
UnsupportedClassVersionError	JDK earlier than 21.	Install JDK 21 and check *java -version*.
payment-service does not start	JWT_SECRET missing, or .NET 8 SDK not installed.	Complete the .env file; install the .NET 8 SDK.
Invalid object name 'payment.payment'	The payment migrations did not run.	Run *docker compose up -d* again and check that payment-migrate ended with code 0.
docker compose build fails: path ../lavarapido-operation-service not found	The operations repository was cloned with another name (for example, *lavarapido-operations-service*).	Rename the folder to *lavarapido-operation-service* or clone it again with that name.

Error or symptom	Probable cause	Solution
Payment screens fail and the other screens work	The payment-service container is not running, or payment-migrate did not end with code 0.	Check *docker compose ps* and *docker compose logs payment-service payment-migrate*.
Liquibase: checksum mismatch	A changeset that already ran was edited.	Revert the change and write a new changeset; in a disposable environment, recreate the database.
Liquibase: waiting for changelog lock	A previous start was interrupted.	Make sure no other instance is running and delete the row of the corresponding DATABASECHANGELOGLOCK_<SERVICE> table.
401 on every request	Expired token or different JWT_SECRET among services.	Log in again; check that all the services read the same .env file.
401 right after signing out in another tab or device	The session was closed and the gateway cuts its token.	Log in again.
Requests take about 3 more seconds and the gateway log says *No se pudo confirmar la sesión*	security-service does not answer; the gateway lets the request through after waiting for it.	Start security-service and check its health.
503 CUSTOMER_SERVICE_UNAVAILABLE or BOOKING_SERVICE_UNAVAILABLE	The service called is down.	Start customer-service or booking-service.
"Your customer profile is not ready yet"	The customer profile has not been created (event not consumed).	Wait and retry; with MESSAGING_ENABLED=true check the customer-service queue.
Emails do not arrive in Mailpit	MAIL_ENABLED=false or wrong host or port.	MAIL_HOST=localhost and MAIL_PORT=1025 on the host; MAIL_DOCKER_HOST=mailpit in Docker.
Booking emails and reminders do not arrive	INTERNAL_API_KEY empty.	Generate the key and restart security-service and notification-service.
Bookings that ended do not appear in "To close", or those without an operator do not become No-show	FOLLOW_UP_ENABLED=false, INTERNAL_API_KEY empty, or booking-service down.	Check the operations-service log (*Could not follow up finished bookings* or *was not closed as not attended*) and the internal key.
Notifications do not arrive	MESSAGING_ENABLED=false or messages in a dead-letter queue.	Enable messaging and review the queues.
CORS error in the browser	The origin is not allowed.	CORS_ALLOWED_ORIGINS=http://localhost:4200,http://localhost:8081 (add the real origin if it is different) and restart the services.
Blank screen or 404 on /api in the web application	The gateway is not running on port 8080.	Start the gateway and check *proxy.conf.json*.
The mobile app does not reach the backend	localhost used from the phone, or different network.	Use 10.0.2.2 in the emulator or the IPv4 address of the computer on the same network.
npm ERESOLVE or build failures	Node.js version not supported by Angular 21.	Use Node.js 22 LTS and reinstall with *npm ci*.
Docker consumes all the memory on Windows	WSL 2 without a memory limit.	Limit the memory in *.wslconfig* and assign at least 5 GB to Docker.
Shifted booking times	Wrong time zone; the system stores UTC.	Check BOOKING_ZONE=America/Bogota.

Note. Prepared by the authors from the infrastructure repository, the runbooks of the services, and the technical backlog.

18. Glossary

Table 15

Glossary of Installation Terms

Term	Definition
API Gateway	Single entry point that routes the requests to the services; it does not check the token signature, but since October 21, 2026 it cuts the tokens of closed sessions.
Changelog	Liquibase file that describes the changes of the database.
Changeset	Unit of change of a changelog, identified by its id and author.
Compose profile	Group of services of Docker Compose that starts only when it is requested (*--profile app*).
Container	Isolated process that runs an application with its dependencies.
CORS	Browser mechanism that restricts requests between different origins.

Term	Definition
Dead-letter queue (DLQ)	Queue where messages that failed after retries are kept.
Exchange	RabbitMQ element that receives messages and routes them to queues.
Health endpoint	Route (/actuator/health) that indicates whether a service is working.
Image	Read-only template from which a container is created.
JDK	Java development kit; the project uses version 21.
Liquibase	Tool for versioning and migrating the database schema.
Mailpit	Development mail server that captures emails without sending them.
Maven Wrapper	Script (*mvnw*) that downloads and runs the correct version of Maven.
Volume	Persistent Docker storage that survives the recreation of a container.
WSL 2	Windows Subsystem for Linux used by Docker Desktop.

Note. Prepared by the authors from the project documentation.

References

Humble, J., & Farley, D. (2010). *Continuous delivery: Reliable software releases through build, test, and deployment automation*. Addison-Wesley.

Microsoft. (n.d.). *Microsoft SQL Server - Ubuntu based images* [Container image]. Docker Hub. Retrieved October 7, 2026, from https://hub.docker.com/r/microsoft/mssql-server

Servicio Nacional de Aprendizaje. (2026). *Guía para elaborar manuales de software: Manual de usuario, manual técnico, manual de instalación y manual de administrador* [Guide for writing software manuals: User manual, technical manual, installation manual, and administrator manual] [Unpublished instructional material]. Technologist in Software Analysis and Development program.

Vehicle Washing Development Team. (2026a). *Vehicle washing management system: Project documentation* (vehicle-w-docs) [Unpublished project documentation]. GitHub. https://github.com/code-sena/vehicle-w-docs

Vehicle Washing Development Team. (2026b). *lavarapido-infra* (develop branch) [Computer software]. GitHub. https://github.com/BrayanCABRERA123/lavarapido-infra

Wiggins, A. (2017). *The twelve-factor app: III. Config*. https://12factor.net/config

19. Annexes

Annex A. Source of the Screenshots

The figures in this manual were taken on October 16, 2026, on a Windows 11 computer with Docker Desktop, following the steps of the manual. The terminal screenshots show the real output of the commands, and the browser screenshot shows the web application running at *http://localhost:4200*. They contain no real passwords, secrets, or personal data: the values of the *.env* file are never shown.

Annex B. Information to Confirm Before Delivery

Table 16

Information Pending Confirmation

Topic	What is missing	Section
One database login per service	The services use sa.	11.4
Variables of October 21, 2026	Resolved on October 22, 2026: the nine of them (*SESSION_CHECK_**, *BOOKING_EXPIRY_**, and *FOLLOW_UP_**) are in *.env.example* and *docker-compose.yml* (action ACO-14, closed).	10.2
Variables of October 24, 2026	*AUTH_RATE_LIMIT_**, *TRUSTED_PROXIES*, *ROUTE_*_TIMEOUT*, and *START_LATE_LIMIT* have a default value but are not yet in *.env.example* or the compose file (ACO-40).	10.2
Published web	Since October 24, 2026 the web image runs nginx without *root* on port 8080 (formerly 80); Caddy points to *web:8080*. Nothing changes locally: *ng serve* stays on 4200.	9
Global database script	*lavarapido-6-services-sqlserver.sql* is not versioned; the database is only created with Liquibase.	11.3

Note. Prepared by the authors from the verification of the repositories and the technical backlog.

Annex C. Summary of Installation Commands

```
# 1. Get the project (repositories side by side; see section 8.1)
git clone -b develop https://github.com/BrayanCABRERA123/lavarapido-infra.git

# 2. Configure the environment
cd lavarapido-infra && cp .env.example .env     # set DB_PASSWORD, JWT_SECRET, INTERNAL_API_KEY

# 3. Infrastructure, database, and payment migrations
docker compose up -d && docker compose ps

# 4. Services (Java and payment-service) and gateway in Docker
docker compose --profile app up -d --build

# 5. Clients
cd ../../Frontend/Front-end-proyecto-web && npm ci && npm start
cd ../LavadoVehicular-Movil && npm ci && npx expo start

# 6. Verification
curl -s http://localhost:8080/actuator/health
```

Annex D. Security Checklist Before Sharing the Environment

Table 17

Environment Security Controls

Control	Met
The .env file is not versioned and has restricted permissions	Yes / No
JWT_SECRET has at least 32 bytes and is not a shared example	Yes / No
The passwords of sa and RabbitMQ and the INTERNAL_API_KEY were changed	Yes / No
The services do not use the sa account to connect	Yes / No
Only the gateway and the web application are exposed outside the machine	Yes / No
Mailpit, the RabbitMQ console, SQL Server, and ports 3001 to 3006 are not accessible from outside	Yes / No
The security service does not run with the dev profile (no demonstration accounts)	Yes / No
HTTPS with TLS 1.2 or later is used if the environment leaves the local machine	Yes / No
Logs do not contain passwords, tokens, or receipts	Yes / No

Note. Prepared by the authors from the security policy of the project.

Annex E. Delivery Checklist

Table 18

Installation Manual Checklist

Criterion	Met
Complete title page and version	Yes / No
Table of contents with updated pages	Yes / No
Objective and scope explained	Yes / No
The commands match the real infrastructure repository	Yes / No
The repository folders keep their original names (including lavarapido-operation-service)	Yes / No
All figures are real screenshots of the installation	Yes / No
Steps verified on a clean installation	Yes / No
No credentials, keys, or sensitive data	Yes / No
Exported to PDF with the name Manual_instalacion_LavadoVehicular_3145556.pdf	Yes / No

Note. Adapted from the checklist of the guide for writing software manuals (SENA, 2026).